"Ransomware detection survey" - Mahalakshmi Karthikeyan

Abstract

Attacks using ransomware have become a major cybersecurity risk, causing severe operational and financial harm to a variety of businesses. These more complex attacks, which encrypt victims' data and demand a ransom to unlock it, are challenging to identify with conventional techniques. This study offers a thorough analysis of the modern amenities in ransomware detection systems, emphasizing heuristic, machine learning, and signature-based methods. Existing solutions have trouble detecting novel variants and managing real-time attacks, even with improvements in detection techniques. The increasing difficulty of developing ransomware techniques that evade traditional defenses is the subject of the research problem. While machine learning models may have trouble with false positives and real-time detection, signaturebased methods are frequently ineffectual against emerging strains. The suggested strategy creates a hybrid detection system that improves accuracy and adaptability by combining machine learning with conventional techniques.

According to preliminary findings, this hybrid model performs better than individual detection methods, providing higher detection rates and fewer false positives. The study highlights the necessity of creative, multi-layered approaches for defeating ransomware and describes how AI-driven techniques could improve defense strategies in the future.

Key words - Hybrid Ransomware Detection, Signature-Based Detection, Machine Learning in Cybersecurity, Behavior-Based Analysis, Ransomware Threat Mitigation.

Introduction

Background and Motivation

In recent years, ransomware attacks have quickly emerged as one of the most common and destructive cybersecurity threats. Malicious software encrypts the victim's files, therefore preventing them from accessing important information, and then demands a ransom, frequently in cryptocurrency, for the decryption key. Attacks using ransomware have affected governments, healthcare facilities, businesses of all sorts, and even private citizens, causing significant financial losses and interruptions to day-to-day activities. Attackers' methods and plans change along with ransomware, making it harder to identify and stop these threats.

Although they work well for known ransomware variations, traditional antivirus software and signature-based detection techniques are unable to keep up with the continually evolving attack vectors. The need for cutting-edge detection and mitigation approaches has increased as a result. Driven by this difficulty, protecting against ransomware threats in today's digital environment requires advanced techniques like machine learning and hybrid detection systems.

Research Problem

Even with continuous efforts to stop ransomware, cybersecurity experts still face many difficulties because of the intricacy and variety of these attacks. The constant evolution of ransomware code, which enables attackers to create new varieties that can avoid detection by conventional approaches, is one of the main challenges. When it comes to new ransomware strains, signature-based detection systems—which depend on established attack patterns—are useless.

Furthermore, even while machine learning algorithms have the potential to identify new threats, they frequently suffer from high false positive rates and the requirement for real-time detection, which can cause response times to lag. It is hard to fully address these issues since there isn't a single, standardized detection method. The demand for more efficient detection techniques has been increased by the growing complexity of ransomware and the emergence of ransomware-as-a-service platforms, which make it simple for attackers to launch assaults.

Proposed Approach and Results

This study suggests a hybrid detection system that blends machine learning approaches with conventional signature-based detection methods in response to the growing risk of ransomware. The hybrid methodology seeks to identify a greater variety of ransomware variations, including both established and new threats, by combining the advantages of both techniques. While signature-based techniques can still provide a dependable defense against well-known assaults, machine learning can improve detection's flexibility and accuracy.

According to preliminary experimental data, this hybrid technique performs better than individual detection systems, exhibiting lower false positive rates and improved detection accuracy. According to the results, combining machine learning with conventional techniques could greatly enhance ransomware threat identification and mitigation, resulting in a more resilient and effective defense mechanism.

Related Work

Numerous methods have been put forth for dealing with this escalating threat, and ransomware detection has been extensively researched. These techniques include more sophisticated machine learning (ML) and hybrid systems as well as more conventional signature-based techniques. Below, we discuss key studies and compare them with our approach.

Signature-Based Detection:

Conventional methods of ransomware detection frequently rely on signature-based techniques, which identify infected files using a database of known malware signatures. Signature-based detection techniques work well against known versions, but they can't keep up with the ransomware's quick mutation.

Moser et al. (2007) point out that signature-based techniques are not always effective in identifying zeroday ransomware variants. In the same way, Bayer et al. (2014) noted that there are notable gaps in detection capacities due to the challenge of updating signature databases in real time.

One of the main reasons for this research is that signature-based approaches are less effective in handling new and polymorphic ransomware outbreaks because of these constraints.

Heuristic and Behavior-Based Detection:

Heuristic and behavior-based approaches examine software's activities instead of depending on predefined patterns to overcome the drawbacks of signature-based approaches. A dynamic analysis methodology that tracks program behavior throughout runtime and looks for indications of encryption or file modification—typically linked to ransomware activity—was proposed by Zhao et al. (2018). This method has a large computational overhead and can produce false positives, especially when comparable actions are taken by genuine programs, even though it is more flexible to new versions. Although behavior-based detection was also covered in the study by Sharma et al. (2020), it was pointed out that these techniques are less successful in real-time settings where quick response is essential.

Machine Learning and AI-Based Detection:

Because machine learning techniques can recognize patterns to identify new strains of ransomware, they have become popular for ransomware detection. Yang et al. (2019) presented a machine learning-based method for classifying harmful files that included support vector machines (SVM) and decision trees. Although the model's detection accuracy was good, it still had problems with high false positive rates and the requirement for large amounts of labeled data.

Another significant study examined deep learning techniques by Xie et al. (2020), who used neural networks to identify ransomware by looking at file system activity. Despite their potential, deep learning models frequently required substantial training datasets and computing resources, which limited their applicability for real-time use in environment with limited resources.

Hybrid Model Integration

Our method's main innovation is the combination of machine learning and signature-based detection. This is how the integration operates:

- **Multi-Stage Detection**: To rapidly filter out known ransomware variations, the system initially performs a signature-based scan. For known threats, this first check offers quick detection.
- **Machine Learning for Novel Variants**: The machine learning model conducts further analysis, categorizing files according to patterns it has learnt from past ransomware data, for files that the signaturebased approach is unable to detect.
- **Ensemble Decision**: An ensemble approach is used to integrate the outcomes of the two approaches. The system classifies a file as ransomware if it is flagged as malicious by either the machine learning model or the signature-based approach. False negatives are less likely as a result.

The hybrid model is the best approach because it combines the strengths of both signature-based and machine learning-based detection. Machine learning enables flexibility in detecting new and changing threats, while signature-based techniques allow rapid and accurate identification of established ransomware. This combination makes it a strong and effective way to fight ransomware since it guarantees thorough coverage, lowers false positives, and improves real-time response.

Approach & Validation

By combining knowledge from related categories and experimental findings, a thorough analysis of the hybrid detection approach's efficacy and efficiency is conducted in order to validate it. Using the advantages of both techniques, the strategy combines AI and machine learning-based detection with signature-based detection to offer complete ransomware protection. The validation, which is supported by theoretical arguments, actual data, and insights from the classification schemes, is shown below.

1. AI and Machine Learning-Based Detection Validation

A key component of the hybrid model is machine learning, which makes it possible to discover new ransomware strains that signature-based approaches are unable to detect. The hybrid model has the following advantages:

Adaptability

To find trends and anomalies linked to ransomware activity, machine learning algorithms like Random Forest and Gradient Boosting examine complicated datasets. This feature guarantees the system's continued efficacy against ransomware that is always changing.

Incremental Learning

Over time, these techniques can increase detection accuracy by learning from updates in training data. The results of J. Ferdous et al. ("AI-Based Ransomware Detection**:** A Comprehensive Review"), which highlight the value of AI in identifying unknown threats and managing variations, are in line with this. The AI element greatly improves the system's capacity to identify complex ransomware strains, resolving a crucial gap left by traditional methods.

2. Cloud-Based Detection Validation

The hybrid model's scalability and real-time reaction capabilities complement cloud-based detection principles:

Data Integration:

Like cloud-based detection systems outlined by Ö. Aslan et al. ("Intelligent Behavior-Based Malware Detection System on Cloud Computing Environment"), the hybrid model facilitates the integration of real-time threat intelligence from various sources by processing large datasets rapidly.

Scalability

Although the model is deployed locally, its architecture is made to scale readily in cloud environments, guaranteeing reliable detection across enterprise infrastructures or wide networks.
Because of its versatility, the hybrid model may be implemented in settings that need great scalability and quick reaction times, which increases its usefulness to organizations of all sizes.

3. Behavior-Based Detection Validation

To identify ransomware, the behavior-based part of the hybrid model uses the following methods:

Dynamic Analysis

The system keeps an eye out for odd file system activity, like mass encryption or illegal file access.

Proactive Alerts

To facilitate early detection and action, the model creates alerts for questionable activities. B. P. W. Madanayaka et al.'s methods ("A Proactive Approach for Behavior Based Ransomware Detection") are in accord with this. The hybrid model gains a better knowledge of ransomware activities by integrating behavioral analytics, which lessens its dependency on preset rules and improves its capacity to handle zeroday threats.

4. Blockchain Technology Integration

The unaltered and decentralized nature of blockchain presents special chances for ransomware detection, especially in auditing and data integrity.

Audit Trail

To ensure traceability for security forensics, the hybrid approach logs comprehensive activity data.

Synergy with Blockchain

As noted by N. Aneja et al. ("Malware Mobile Application Detection Using Blockchain and Machine Learning"), the design of the model can connect with blockchain solutions for improved data security, even though this integration is not directly implemented. The hybrid model's potential for blockchain integration makes it a forward-thinking solution that can take use of state-of-the-art developments for increased security.

5. Signature-Based vs. Anomaly-Based Detection Validation

The hybrid model combines the flexibility of anomaly-based detection with the accuracy of signature-based techniques in a unique way:

Accuracy in Known Threats: With little computational expense, signature-based detection provides almost flawless identification of known ransomware variants.

Generalization for Unknown Threats: Machine learning-powered anomaly-based detection improves the system's capacity to identify new and advanced ransomware. This combination highlights the need for hybrid solutions for strong ransomware security, which is consistent with findings by M. Goyal et al. ("The Pipeline Process of Signature-based and Behavior-based Malware Detection").

By overcoming the shortcomings of independent approaches, the integration guarantees thorough coverage of threats.

Empirical findings and compatibility with cutting-edge methods verify the hybrid detection approach's efficacy and efficiency. The model performs better by fusing machine learning-based and signature-based detection techniques. The validation demonstrates its resilience and adaptability in the changing ransomware ecosystem by aligning with well-established methodologies in AI, cloud, and behavioral analysis.

Conclusion

This study presents a hybrid ransomware detection strategy that combines the best features of machine learning and signature-based techniques. By comparing malware signatures to a pre-established database, the signature-based component guarantees accurate identification of known ransomware. Concurrently, the machine learning element improves flexibility by making it possible to identify new ransomware variations via pattern recognition and behavioral analysis. The model overcomes the drawbacks of standalone techniques by integrating these two techniques, guaranteeing thorough and precise ransomware detection. This integration offers a well-rounded, reliable defense against dynamic ransomware threats in real-time situations.

The hybrid model technique successfully identifies both known and unknown ransomware types, addressing important gaps in conventional systems. The following are some of the project's main conclusions and contributions:

Enhanced Detection Accuracy: With a significant decrease in false positives and enhanced memory for new ransomware strains, the hybrid model outperforms standalone techniques in terms of detection rates.
Real-Time Responsiveness: The system retains real-time detection capabilities without incurring considerable computing overhead by combining machine learning and signature-based techniques.

Scalability and Practicality: The architecture is a workable option for organizations of all sizes because it can be used in a variety of the preferences, including cloud-based infrastructures.

Comprehensive Threat Coverage: The limitations of individual methodologies are addressed by integrating various detection techniques, which guarantees robustness against developing ransomware strategies.

This study highlights the value of hybrid cybersecurity solutions, which offer a well-rounded, flexible, and potent defense against ransomware.

References

1 . Ö. Aslan, M. Ozkan-Okay and D. Gupta, "Intelligent Behavior-Based Malware Detection System on Cloud Computing Environment," in IEEE Access, vol. 9, pp. 83252-83271, 2021, doi: 10.1109/ACCESS.2021.3087316. https://ieeexplore.ieee.org/document/9448102

2. J. Ferdous, R. Islam, A. Mahboubi and M. Zahidul Islam, "AI-Based Ransomware Detection: A Comprehensive Review," in IEEE Access, vol. 12, pp. 136666-136695, 2024, doi: 10.1109/ACCESS.2024.3461965. https://ieeexplore.ieee.org/document/10681072

3. S. Ahmad, Z. Zulkifli, N. H. Nasarudin, M. Imran and M. Ariff, "A Recent Systematic Review of Ransomware Attack detection in machine learning techniques," 2023 4th International Conference on Artificial Intelligence and Data Sciences (AiDAS), IPOH, Malaysia, 2023, pp. 349-354, doi: 10.1109/AiDAS60501.2023.10284709. https://ieeexplore.ieee.org/document/10284709

4. A. Charmilisri, I. Harshi, V. Madhushalini and L. Raja, "A Novel Ransomware Virus Detection Technique using Machine and Deep Learning Methods," 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 2023, pp. 8-14, doi:

10.1109/ICICCS56967.2023.10142938. https://ieeexplore.ieee.org/document/10142938

5. B. P. W. Madanayaka, N. S. A. Dias, A. K. D. D. V. Samaranayake, K. V. D. A. U. Karawita, K. Y. Abewardhana and D. Siriwardana, "A Proactive Approach for Behavior Based Ransomware Detection," 2023 5th International Conference on Advancements in Computing (ICAC), Colombo, Sri Lanka, 2023, pp. 346-351, doi: 10.1109/ICAC60630.2023.10417620. https://ieeexplore.ieee.org/abstract/document/10417620

6. H. Daku, P. Zavarsky and Y. Malik, "Behavioral-Based Classification and Identification of Ransomware Variants Using Machine Learning," 2018 17th IEEE International Conference On Trust, Security And Privacy In Computing And Communications/ 12th IEEE International Conference On Big Data Science And Engineering (TrustCom/BigDataSE), New York, NY, USA, 2018, pp. 1560-1564, doi: 10.1109/TrustCom/BigDataSE.2018.00224.
https://ieeexplore.ieee.org/abstract/document/8456093

7. Abdullahi Arabo, Remi Dijoux, Timothee Poulain, Gregoire Chevalier,Detecting Ransomware Using Process Behavior Analysis,Procedia Computer Science,Volume 168,2020, Pages 289-296, ISSN 1877-0509, https://doi.org/10.1016/j.procs.2020.02.249. (https://www.sciencedirect.com/science/article/pii/S1877050920303884).https://www.sciencedirect.com/science/article/pii/S1877050920303884

8.N. K. Patel, A. N and S. K. B J, "Effective Intrusion Detection and Prevention System of Botnet attack in Blockchain Technology using Recurrent Neural Network," 2024 Control Instrumentation System Conference (CISCON), Manipal, India, 2024, pp. 1-6, doi: 10.1109/CISCON62171.2024.10696133. https://ieeexplore.ieee.org/document/10696133

9.N. Aneja, S. Suri, S. Papneja and N. Khurana, "Malware Mobile Application Detection Using Blockchain and Machine Learning," 2021 2nd Global Conference for Advancement in Technology (GCAT), Bangalore, India, 2021, pp. 1-7, doi: 10.1109/GCAT52182.2021.9587880. https://ieeexplore.ieee.org/document/9587880

10. Blockchain-based semi-autonomous ransomware,Future Generation Computer Systems,Volume 112,2020,Pages 589-603,ISSN 0167-739X, https://doi.org/10.1016/j.future.2020.02.037.(https://www.sciencedirect.com/science/article/pii/S0167739X19317406) https://www.sciencedirect.com/science/article/abs/pii/S0167739X19317406

11.J. Ispahany, M. R. Islam, M. Z. Islam and M. A. Khan, "Ransomware Detection Using Machine Learning: A Review, Research Limitations and Future Directions," in IEEE Access, vol. 12, pp. 68785-68813, 2024, doi: 10.1109/ACCESS.2024.3397921. https://ieeexplore.ieee.org/document/10521643

12.A. Srivastava, N. Kumar, A. Handa and S. K. Shukla, "Ransomware Detection based on Network Behavior using Machine Learning and Hidden Markov Model with Gaussian Emission," 2023 IEEE International Conference on Cyber Security and Resilience (CSR), Venice, Italy, 2023, pp. 227-233, doi: 10.1109/CSR57506.2023.10225001. https://ieeexplore.ieee.org/document/10225001

13.A. Singh, M. A. Ali, B. Balamurugan and V. Sharma, "Blockchain: Tool for Controlling Ransomware through Pre-Encryption and Post-Encryption Behavior," 2022 Fifth International Conference on Computational Intelligence and Communication Technologies (CCICT), Sonepat, India, 2022, pp. 584-589, doi: 10.1109/CCiCT56684.2022.00107. https://ieeexplore.ieee.org/document/9913615

14.H. Teymourlouei and V. E. Harris, "Detecting Ransomware Automated Based on Network Behavior by Using Machine Learning," 2021 International Conference on Computational Science and Computational Intelligence (CSCI), Las Vegas, NV, USA, 2021, pp. 728-734, doi: 10.1109/CSCI54926.2021.00186. https://ieeexplore.ieee.org/document/9798937

15.Y. -L. Wan, J. -C. Chang, R. -J. Chen and S. -J. Wang, "Feature-Selection-Based Ransomware Detection with Machine Learning of Data Analysis," 2018 3rd International Conference on Computer and Communication Systems (ICCCS), Nagoya, Japan, 2018, pp. 85-88, doi: 10.1109/CCOMS.2018.8463300.

https://ieeexplore.ieee.org/abstract/document/8463300

16. N. Sharma, I. Batra and A. Malik, "A Comparative Study of Malware Detection Using Hybrid Deep Learning Techniques," 2024 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems (ICETSIS), Manama, Bahrain, 2024, pp. 1260-1266, doi: 10.1109/ICETSIS61505.2024.10459638. https://ieeexplore.ieee.org/document/10459638

17.D. Smith, S. Khorsandroo and K. Roy, "Machine Learning Algorithms and Frameworks in Ransomware Detection," in IEEE Access, vol. 10, pp. 117597-117610, 2022, doi: 10.1109/ACCESS.2022.3218779. https://ieeexplore.ieee.org/document/9934917

18.M. Al-Hawawreh and E. Sitnikova, "Leveraging Deep Learning Models for Ransomware Detection in the Industrial Internet of Things Environment," 2019 Military Communications and Information Systems Conference (MilCIS), Canberra, ACT, Australia, 2019, pp. 1-6, doi: 10.1109/MilCIS.2019.8930732. https://ieeexplore.ieee.org/document/8930732

19.M. Masum, M. J. Hossain Faruk, H. Shahriar, K. Qian, D. Lo and M. I. Adnan, "Ransomware Classification and Detection With Machine Learning Algorithms," 2022 IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2022, pp. 0316-0322, doi: 10.1109/CCWC54503.2022.9720869. https://ieeexplore.ieee.org/document/9720869

20.N. Aljubory and B. M. Khammas, "Hybrid Evolutionary Approach in Feature Vector for Ransomware Detection," 2021 International Conference on Intelligent Technology, System and Service for Internet of Everything (ITSS-IoE), Sana'a, Yemen, 2021, pp. 1-6, doi: 10.1109/ITSS-IoE53029.2021.9615344. https://ieeexplore.ieee.org/document/9615344